



US 20250275582A1

(19) **United States**

(12) **Patent Application Publication**
BELL et al.

(10) **Pub. No.: US 2025/0275582 A1**

(43) **Pub. Date: Sep. 4, 2025**

(54) **AN AEROSOL PROVISION SYSTEM**

Publication Classification

(71) Applicant: **NICOVENTURES TRADING LIMITED**, London (GB)

(51) **Int. Cl.**

A24F 40/51 (2020.01)

A24F 7/00 (2006.01)

A24F 40/40 (2020.01)

A24F 40/70 (2020.01)

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(52) **U.S. Cl.**

CPC *A24F 40/51* (2020.01); *A24F 7/00* (2013.01); *A24F 40/40* (2020.01); *A24F 40/70* (2020.01)

(21) Appl. No.: **18/858,007**

(22) PCT Filed: **Apr. 20, 2023**

(86) PCT No.: **PCT/GB2023/051036**

§ 371 (c)(1),

(2) Date: **Oct. 18, 2024**

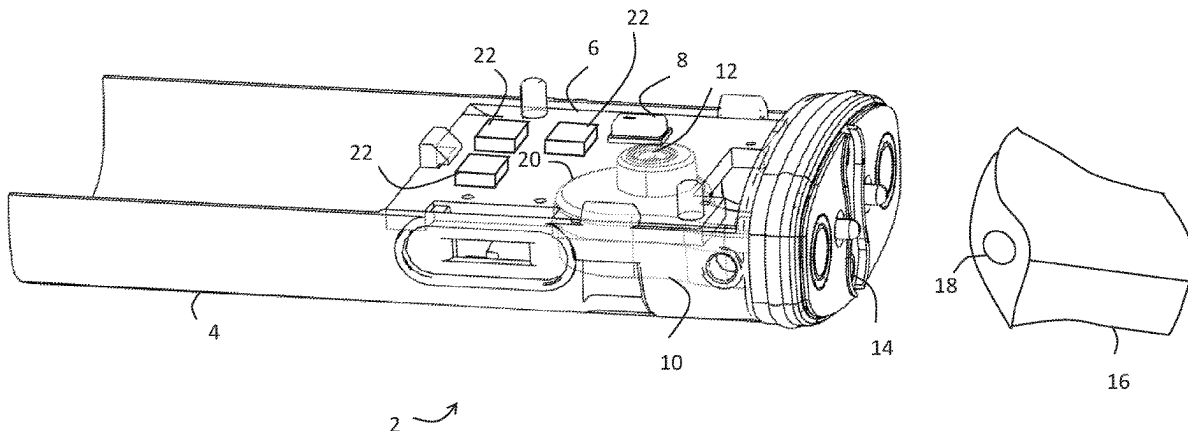
(30) **Foreign Application Priority Data**

Apr. 22, 2022 (CN) 2022104279884

(57)

ABSTRACT

An aerosol provision system has a pressure sensing apparatus including a circuit board and a pressure sensor. A first side of the pressure sensor is in fluid communication with a mouthpiece of the aerosol provision system and a second side of the pressure sensor is in fluid communication with ambient air. The pressure sensor is mounted on the circuit board to provide the fluid communication to the first side or the second side. The circuit board has a hole to provide the fluid communication to the other of the first side and the second side of the pressure sensor.



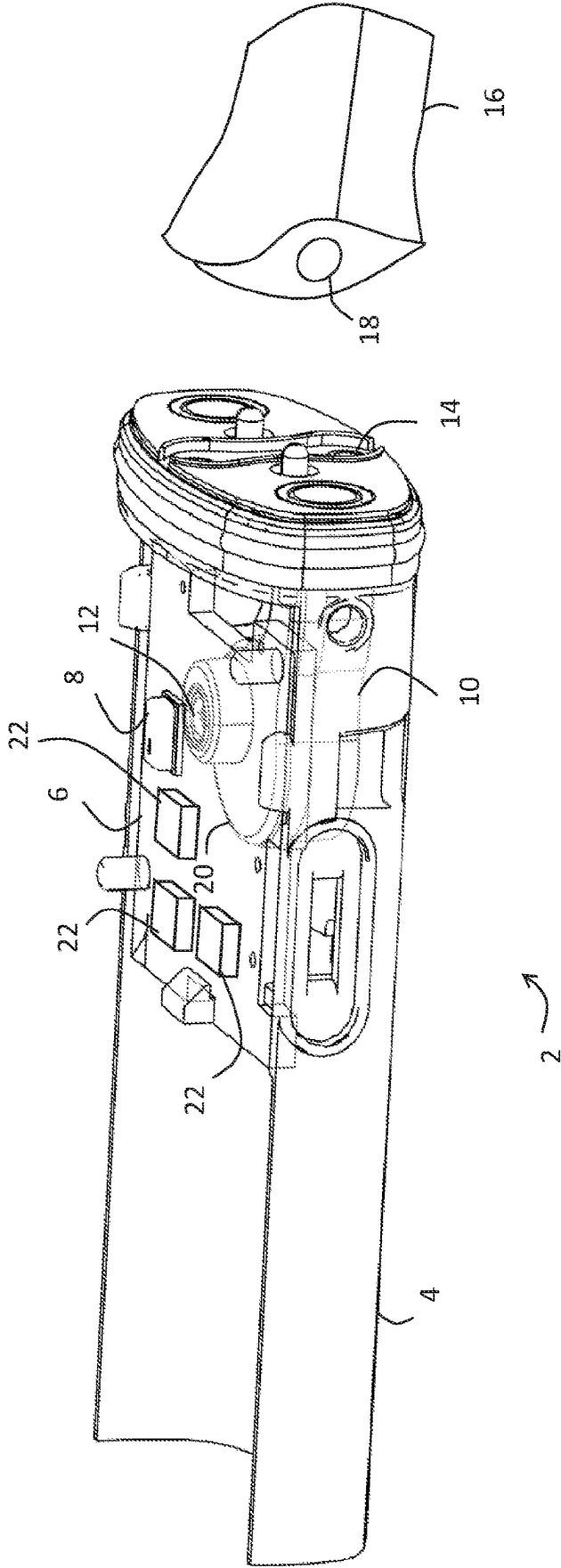


FIG. 1

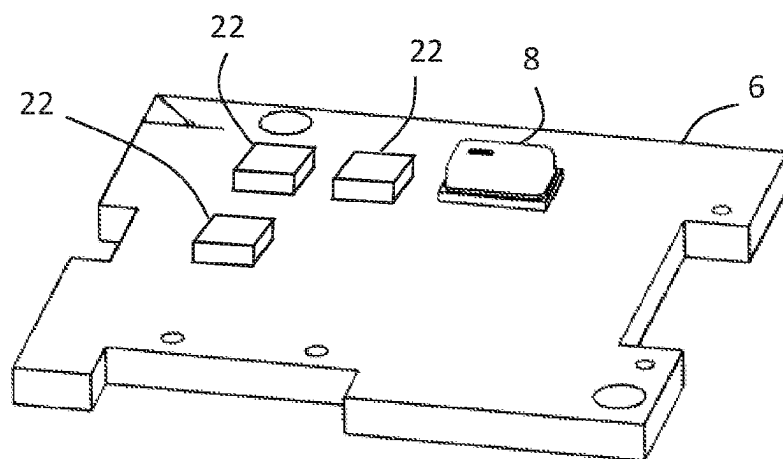


FIG. 2

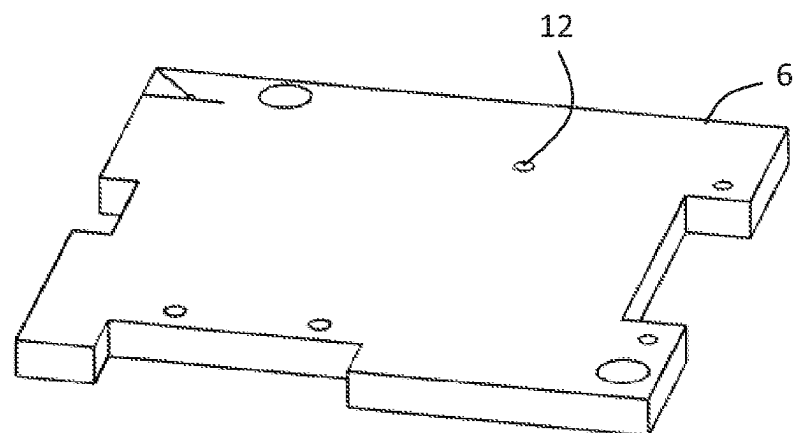


FIG. 3

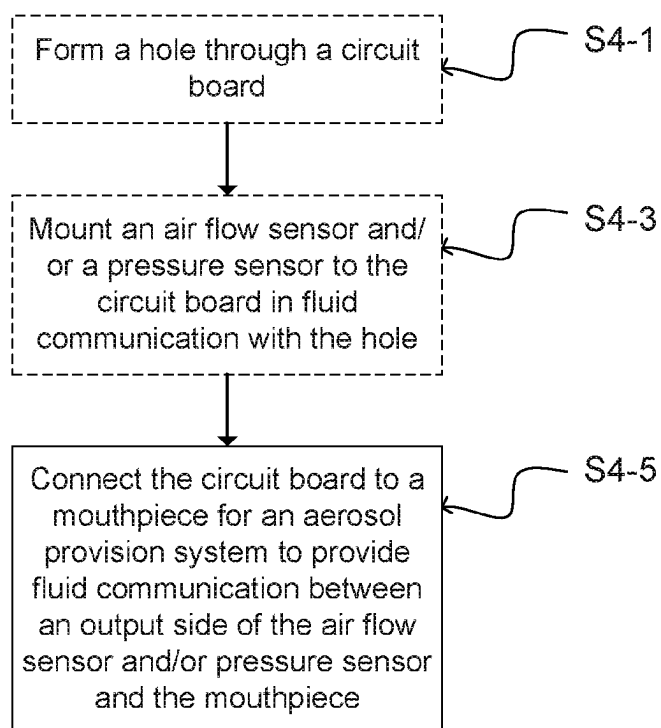


FIG 4

AN AEROSOL PROVISION SYSTEM

RELATED APPLICATIONS

[0001] The present application is a National Phase entry of PCT Application No. PCT/GB2023/051036 filed Apr. 20, 2023, which claims priority to CN application No. 2022104279884 filed Apr. 22, 2022, each of which is hereby incorporated by reference in their entirety.

FIELD AND BACKGROUND

[0002] The present disclosure relates to the field of aerosol provision systems. In particular, but not exclusively, the present disclosure relates to an aerosol provision system with an air flow and/or pressure sensing apparatus.

[0003] Known approaches are described in WO2017139662A1, WO2018172765A1, EP3560360A1, WO2020157634A1, and WO2019122866A1.

SUMMARY

[0004] Viewed from a first aspect, there is provided an aerosol provision system comprising: an air flow and/or pressure sensing apparatus comprising a circuit board and an air flow sensor and/or a pressure sensor; wherein a first side of the air flow sensor and/or pressure sensor is in fluid communication with a mouthpiece of the aerosol provision system and a second side of the air flow sensor and/or pressure sensor is in fluid communication with ambient air; the air flow sensor and/or pressure sensor is mounted on the circuit board to provide the fluid communication to the first side or the second side; and the circuit board has a hole to provide the fluid communication to the other of the first side and the second side of the air flow sensor and/or pressure sensor.

[0005] Viewed from a second aspect, there is provided a method of manufacturing an aerosol provision system, the method comprising: connecting a circuit board to a mouthpiece for the aerosol provision system, wherein the circuit board comprises an air flow sensor and/or a pressure sensor and wherein an input side or an output side of the air flow sensor and/or pressure sensor is mounted to a hole through the circuit board, and wherein after the connecting the output side of the air flow sensor and/or pressure sensor is in fluid communication with the mouthpiece of the aerosol provision system, and the input side of the air flow sensor and/or pressure sensor is in fluid communication with ambient air.

BRIEF DESCRIPTION OF FIGURES

[0006] Embodiments and examples of the present approaches will now be described, by way of example only, with reference to the accompanying drawings, in which:

[0007] FIG. 1 is a schematic of an aerosol provision system in accordance with the techniques described herein;

[0008] FIG. 2 illustrates a circuit board and an airflow and/or pressure sensor of the aerosol provision system of FIG. 1;

[0009] FIG. 3 illustrates the circuit board with the airflow and/or pressure sensor removed so that a hole provided in the circuit board is visible; and

[0010] FIG. 4 is a flowchart depicting a method of manufacturing an aerosol provision system.

[0011] While the presently described approach is susceptible to various modifications and alternative forms, specific embodiments are shown by way of example in the drawings

and are herein described in detail. It should be understood, however, that drawings and detailed description thereto are not intended to limit the scope to the particular form disclosed, but on the contrary, the scope is to cover all modifications, equivalents and alternatives falling within the spirit and scope as defined by the appended claims.

DETAILED DESCRIPTION

[0012] According to the present disclosure, a “non-combustible” aerosol provision system is an aerosol provision system where a constituent aerosol-generating material of the aerosol provision system (or component thereof) is not combusted or burned in order to facilitate delivery of at least one substance to a user.

[0013] The non-combustible aerosol provision system may be an electronic cigarette, also known as a vaping device or electronic nicotine delivery system (END), although it is noted that the presence of nicotine in the aerosol-generating material is not a requirement.

[0014] The non-combustible aerosol provision system may be an aerosol-generating material heating system, also known as a heat-not-burn system. An example of such a system is a tobacco heating system.

[0015] The non-combustible aerosol provision system may be a hybrid system to generate aerosol using a combination of aerosol-generating materials, one or a plurality of which may be heated. Each of the aerosol-generating materials may be, for example, in the form of a solid, liquid or gel and may or may not contain nicotine. The hybrid system may comprise a liquid or gel aerosol-generating material and a solid aerosol-generating material. The solid aerosol-generating material may comprise, for example, tobacco or a non-tobacco product.

[0016] Typically, the non-combustible aerosol provision system may comprise a non-combustible aerosol provision device and a consumable for use with the non-combustible aerosol provision device.

[0017] The non-combustible aerosol provision system, such as a non-combustible aerosol provision device thereof, may comprise a power source and a controller. The power source may, for example, be an electric power source or an exothermic power source. The exothermic power source comprises a carbon substrate which may be energised so as to distribute power in the form of heat to an aerosol-generating material or to a heat transfer material in proximity to the exothermic power source.

[0018] The non-combustible aerosol provision system may comprise an area for receiving the consumable, an aerosol generator, an aerosol generation area, a housing, a mouthpiece, a filter and/or an aerosol-modifying agent.

[0019] The consumable for use with the non-combustible aerosol provision device may comprise aerosol-generating material, an aerosol-generating material storage area, an aerosol-generating material transfer component, an aerosol generator, an aerosol generation area, a housing, a wrapper, a filter, a mouthpiece, and/or an aerosol-modifying agent.

[0020] A consumable is an article comprising or consisting of aerosol-generating material, part or all of which is intended to be consumed during use by a user. A consumable may comprise one or more other components, such as an aerosol-generating material storage area, an aerosol-generating material transfer component, an aerosol generation area, a housing, a wrapper, a mouthpiece, a filter and/or an aerosol-modifying agent. A consumable may also comprise

an aerosol generator, such as a heater, that emits heat to cause the aerosol-generating material to generate aerosol in use. The heater may, for example, comprise combustible material, a material heatable by electrical conduction, or a susceptor.

[0021] A susceptor is a material that is heatable by penetration with a varying magnetic field, such as an alternating magnetic field. The susceptor may be an electrically-conductive material, so that penetration thereof with a varying magnetic field causes induction heating of the heating material. The heating material may be magnetic material, so that penetration thereof with a varying magnetic field causes magnetic hysteresis heating of the heating material. The susceptor may be both electrically-conductive and magnetic, so that the susceptor is heatable by both heating mechanisms. The device that is configured to generate the varying magnetic field is referred to as a magnetic field generator, herein.

[0022] An aerosol generator is an apparatus configured to cause aerosol to be generated from the aerosol-generating material. In some embodiments, the aerosol generator is a heater configured to subject the aerosol-generating material to heat energy, so as to release one or more volatiles from the aerosol-generating material to form an aerosol. In some embodiments, the aerosol generator is configured to cause an aerosol to be generated from the aerosol-generating material without heating. For example, the aerosol generator may be configured to subject the aerosol-generating material to one or more of vibration, increased pressure, or electrostatic energy.

[0023] An aerosol provision system may be provided with an air flow sensor and/or a pressure sensor to measure an airflow through and/or pressure within the aerosol provision system. For example, the air flow sensor and/or pressure sensor may be used to detect the pressure of a user's puff on the aerosol provision system and/or detect a flow of air through the aerosol provision system caused by a user's puff. The pressure may be measured as an absolute pressure or as a relative pressure relative to an ambient pressure. This ability to sense a pressure of, and/or airflow caused by, a user's puff may be used to identify when a user begins puffing on the aerosol provision system and/or to measure the strength of the user's puff. These air flow and/or pressure readings may be stored and provided to the user, or may be used to control the operation aerosol provision system for example. In some examples, the aerosol provision system begins generating aerosols in response to detecting a puff by the user on the basis of the air flow and/or pressure measured by the air flow sensor and/or pressure sensor.

[0024] One approach to providing an air flow sensor and/or a pressure sensor in an aerosol provision system would be to mount an air flow sensor and/or pressure sensor on a circuit board of the aerosol provision system along with other electronic components that are provided. With the air flow sensor and/or pressure sensor mounted in this way, a passage could be provided from a mouthpiece of the aerosol provision system to place the mouthpiece and the airflow and/or pressure sensor in fluid communication such that the air flow and/or pressure experienced at the air flow sensor and/or pressure sensor is the same as that imparted at the mouthpiece by virtue of the user's puff.

[0025] However, with such an approach the aerosol provision system may be complicated to manufacture and considerable constraints may be placed on the design of the

aerosol provision system in order to be able to ensure adequate fluid communication between the air flow sensor and/or pressure sensor and the mouthpiece. Specifically, with the air flow sensor and/or pressure sensor mounted along with the other electronic components on the circuit board, it may be difficult to provide a fluid path to the air flow sensor and/or pressure sensor due to the presence of those other components. Moreover, in order to ensure that the airflow and/or pressure experienced at the air flow sensor and/or pressure sensor is the same as the airflow and/or pressure at the mouthpiece, a seal may be provided around the air flow sensor and/or pressure sensor to avoid dissipation of air into a housing of the aerosol provision system, thereby reducing the air flow and/or pressure measured at the air flow sensor and/or pressure sensor. The provision of this seal is made difficult due to the presence of the other electrical components on the circuit board and valuable area on the circuit board may be occupied by the seal. This difficulty is exacerbated by the need to provide a seal that is larger than the air flow sensor and/or pressure sensor itself so as to provide a seal around the air flow sensor and/or pressure sensor. This approach therefore may lead to a bulky arrangement within the aerosol provision system that is difficult to manufacture.

[0026] In accordance with the techniques described herein however, there is provided an aerosol provision system with an air flow and/or pressure sensing apparatus including a circuit board through which a hole is provided to allow fluid communication to the air flow sensor and/or pressure sensor. With this hole provided, the air flow sensor and/or pressure sensor may be placed in fluid communication with either the ambient air or the mouthpiece via a fluid path that passes through the hole. This provides significant additional flexibility as to how the components within the aerosol provision system can be arranged so as to provide the desired fluid paths.

[0027] For example, instead of providing a seal around the air flow sensor and/or pressure sensor itself which would potentially interfere with other components on the circuit board and which may need to be larger than the air flow sensor and/or pressure sensor itself in order to provide an adequate seal, the seal may be provided on the other side of the circuit board. By providing a seal on the opposite side of the circuit board around the hole, a smaller seal can be used and a fluid path to the mouthpiece may be easier to provide within the internal space available in the aerosol provision system.

[0028] Particular examples will now be described with reference to the figures.

[0029] FIG. 1 shows an aerosol provision system 2 having a housing 4 in which a circuit board 6 is arranged. The circuit board 6 has an air flow sensor and/or pressure sensor 8 and various other electrical components 22 mounted on a component mounting side thereof, where the components 22 may for example comprise a microprocessor, storage, and one or more communication modules (e.g., a Bluetooth™ module).

[0030] A first side of the air flow sensor and/or pressure sensor 8 (which in FIG. 1 is nominally depicted as the top side of the air flow sensor and/or pressure sensor 8) is exposed to the inside of the housing 4 and from there to the external air. As such, the first side the air flow sensor and/or pressure sensor 8 is in fluid communication with ambient air and so is at ambient pressure.

[0031] Where reference is made to relative positions of features within the figures, the positioning referred to is defined relative to a nominal orientation of the aerosol provision system 2 and its components as depicted in the figures. It will be appreciated that if the aerosol provision system 2 were to be reoriented, the orientation of the components may correspondingly vary.

[0032] Below the air flow sensor and/or pressure sensor 8 as depicted in FIG. 1 is provided a hole 12 in the circuit board 6. A passage 10 is provided to connect the underside of the hole 12 and a first aperture 14. When a mouthpiece 16 is affixed to the housing 4, the first aperture 14 corresponds with a second aperture 18 provided in the mouthpiece 16 so as to provide a fluid path from a second side of the air flow sensor and/or pressure sensor 8 (depicted in FIG. 1 on the underside of the air flow sensor and/or pressure sensor 8) through the hole 12, along the passage 10, through the first aperture 14 and second aperture 18 to the mouthpiece 16. In this way, the second side of the air flow sensor and/or pressure sensor 8 is placed in fluid communication with the mouthpiece 16. A seal 20 is provided around the hole 12 on a non-mounting side (shown in FIG. 1 as the underside) of the circuit board 6 in order to seal the passage 10.

[0033] With this arrangement, when the user puffs on the mouthpiece 16 of the aerosol provision system 2 (which may be a non-combustible aerosol provision system), the air flow and/or pressure in the mouthpiece 16 due to the user's puff is also experienced at the second side of the air flow sensor and/or pressure sensor 8, thereby allowing the air flow sensor and/or pressure sensor 8 to measure the air flow and/or pressure of the puff.

[0034] By providing the hole 8, the seal 20 can be formed on the opposite side of the circuit board 6 to the side on which the air flow sensor and/or pressure sensor 8 and the other electrical components are mounted. The seal 20 can therefore be placed such that it does not occupy area on the mounting side of circuit board 6 which could otherwise be used to house components 22. Further, since the seal 20 in the example of FIG. 1 seals the hole 12 rather than providing a seal around the air flow sensor and/or pressure sensor 8, the seal can be made smaller than would otherwise be the case if the seal had to be provided around the air flow sensor and/or pressure sensor 8. For example, the width of the seal 20 can be made smaller than the width of the air flow sensor and/or pressure sensor 8. This reduces the amount of space occupied inside the housing 4 to provide the fluid path between the mouthpiece 16 and the air flow sensor and/or pressure sensor 8.

[0035] In some examples, the air flow sensor and/or pressure sensor 8 is a differential pressure sensor arranged to output a signal indicative of a difference in pressure between the air in the mouthpiece 16 and the ambient air. The arrangement discussed above means that the provision of ambient air to the first side of the pressure sensor 8 and the provision of a fluid path to the mouthpiece 16 from the other side of the pressure sensor 8 can be simplified. As such, this arrangement simplifies the use of a differential pressure sensor in the aerosol provision system 2. Use of a differential pressure sensor may be appropriate since a differential pressure from a differential pressure sensor may provide a more accurate indicator of a user's puff strength than an absolute pressure.

[0036] In some examples, the air flow sensor and/or pressure sensor 8 is a micro electro-mechanical system (MEMS)

sensor. A MEMS sensor consumes little power in operation and can respond to small changes in air flow and/or pressure and so may be particularly appropriate in a portable device such as an aerosol provision system

[0037] FIG. 2 shows an air flow and/or pressure sensing apparatus comprising the circuit board 6, air flow sensor and/or pressure sensor 8, and various other components 22 as shown in FIG. 1. The air flow sensor and/or pressure sensor 8 is mounted such that the second side of the air flow sensor and/or pressure sensor 8 is in fluid communication with a hole 12 in the circuit board 6 provided underneath (according to the orientation shown in Figure) the air flow sensor and/or pressure sensor 8 (not visible in FIG. 2) and the first side (i.e., the top side in FIG. 2) of the air flow sensor and/or pressure sensor 8 is in fluid communication with ambient air.

[0038] FIG. 3 illustrates the same circuit board 6 with the air flow sensor and/or pressure sensor 8 and the other components 22 removed. With the air flow sensor and/or pressure sensor 8 removed, the hole 12 can be seen. As can be seen from FIG. 3, the circuit board 6 may be provided with further holes which may be used, for example, for mounting the circuit board 6. Conventional techniques for providing holes within a circuit board may be used to produce the hole 12 above which the air flow sensor and/or pressure sensor 8 is mounted. As such, a dedicated process need not be used to provide the hole 12, allowing the hole 12 to be made easily and without incurring extensive additional cost. The techniques described herein may be carried out with a hole 12 that is plated or unplated.

[0039] FIG. 4 is a flowchart illustrating a method of manufacturing an aerosol provision system. At step S4-1, a hole is made in the circuit board 6. This can be made using conventional techniques for making holes in circuit board e.g., for making mounting holes. This step is indicated as optional as a manufacture method for an aerosol provision system may commence with a circuit board that already includes a suitable hole.

[0040] At step S4-1, an air flow sensor and/or pressure sensor 8 is mounted on the circuit board in fluid communication with the hole. As is made clear above, the air flow sensor and/or pressure sensor may be mounted on either side of the board, i.e. the air flow sensor and/or pressure sensor may be mounted on a component-mounting side of the board with an outlet side of the air flow sensor and/or pressure sensor directed to be in fluid communication with the hole, or the air flow sensor and/or pressure sensor may be mounted on the other side of the board with the inlet side of the air flow sensor and/or pressure sensor directed to be in fluid communication with the hole. This step is indicated as optional as a manufacture method for an aerosol provision system may commence with an air flow sensor and/or pressure sensor already mounted to the circuit board.

[0041] At step S4-5, the circuit board is connected to a mouthpiece for the aerosol provision system, to provide fluid communication between the output side of the air flow sensor and/or pressure sensor and the mouthpiece. As explained above, a seal may be provided to seal between the circuit board (for example if the air flow sensor and/or pressure sensor is mounted with the output side in fluid communication with the hole) and an air passage to the mouthpiece. Likewise, if the air flow sensor and/or pressure sensor is mounted with the input side in fluid communication with the hole, then such a seal may seal the circuit board

around the air flow sensor and/or pressure sensor, and/or seal to the air flow sensor and/or pressure sensor itself. In manufacturing terms, such a seal may be attached to the circuit board and/or to the air flow sensor and/or pressure sensor before being attached to the mouthpiece (or an air passage thereto), or vice versa.

[0042] Thus, there has been described an aerosol provision system and a method of manufacturing an aerosol provision system that can allow an air flow sensor and/or pressure sensor to be more easily provided within the aerosol provision system. By making a hole in the circuit board through which air can flow, additional flexibility as to how the air flow sensor and/or pressure sensor is connected to a mouthpiece of the aerosol provision system can be provided and the internal design of the aerosol provision system can be simplified.

[0043] In the present application, the words “configured to . . .” are used to mean that an element of an apparatus has a configuration able to carry out the defined operation. In this context, a “configuration” means an arrangement or manner of interconnection of hardware or software. For example, the apparatus may have dedicated hardware which provides the defined operation, or a processor or other processing device may be programmed to perform the function. “Configured to” does not imply that the apparatus element needs to be changed in any way in order to provide the defined operation.

[0044] The various embodiments described herein are presented only to assist in understanding and teaching the claimed features. These embodiments are provided as a representative sample of embodiments only, and are not exhaustive and/or exclusive. It is to be understood that advantages, embodiments, examples, functions, features, structures, and/or other aspects described herein are not to be considered limitations on the scope of the invention as defined by the claims or limitations on equivalents to the claims, and that other embodiments may be utilized and modifications may be made without departing from the scope of the claimed invention. Various embodiments of the invention may suitably comprise, consist of, or consist essentially of, appropriate combinations of the disclosed elements, components, features, parts, steps, means, etc., other than those specifically described herein. In addition, this disclosure may include other inventions not presently claimed, but which may be claimed in future.

1. An aerosol provision system comprising:
 - an air flow and/or pressure sensing apparatus comprising a circuit board and an air flow sensor and/or a pressure sensor;
 - wherein a first side of the air flow sensor and/or pressure sensor is in fluid communication with a mouthpiece of the aerosol provision system and a second side of the air flow sensor and/or pressure sensor is in fluid communication with ambient air;
 - the air flow sensor and/or pressure sensor is mounted on the circuit board to provide the fluid communication to the first side or the second side; and
 - the circuit board has a hole to provide the fluid communication to the other of the first side and the second side of the air flow sensor and/or pressure sensor.
2. The aerosol provision system according to claim 1, wherein:
 - the air flow sensor and/or pressure sensor is a differential pressure sensor configured to measure a pressure dif-

ference between a pressure on the first side of the pressure sensor and a pressure on the second side of the pressure sensor.

3. The aerosol provision system according to claim 1, wherein:
 - the air flow sensor and/or pressure sensor is a micro electro-mechanical system (MEMS) sensor.
4. The aerosol provision system according to claim 1, the aerosol provision system further comprising:
 - a seal positioned on an opposite side of the circuit board to the air flow sensor and/or pressure sensor, and providing a seal around the hole.
5. The aerosol provision system according to claim 4, wherein:
 - a width of the seal is smaller than a width of the air flow sensor and/or pressure sensor.
6. A method of manufacturing an aerosol provision system, the method comprising:
 - connecting a circuit board to a mouthpiece for the aerosol provision system, wherein the circuit board comprises an air flow sensor and/or a pressure sensor and wherein an input side or an output side of the air flow sensor and/or pressure sensor is mounted to a hole through the circuit board, and wherein after the connecting the output side of the air flow sensor and/or pressure sensor is in fluid communication with the mouthpiece of the aerosol provision system, and the input side of the air flow sensor and/or pressure sensor is in fluid communication with ambient air.
7. The method of claim 6, wherein the connecting further comprising connecting the output side of the air flow sensor and/or pressure sensor to the mouthpiece via a seal that seals an air passage to the output side of the air flow sensor and/or pressure sensor.
8. The method of claim 7, wherein the circuit board comprises a plurality of electrical components attached to a first side thereof, and wherein the seal is provided on a second side of the circuit board.
9. The method of claim 7, wherein the seal is attached to the circuit board prior to the connecting.
10. The method of claim 7, wherein the air passage connects the output side of the air flow sensor and/or pressure sensor to the mouthpiece via the hole through the circuit board.
11. The method according to claim 7, wherein a width of the seal is smaller than a width of the air flow sensor and/or pressure sensor.
12. The method of claim 6, further comprising, prior to the connecting:
 - mounting the air flow sensor and/or pressure sensor to the circuit board.
13. The method of claim 12, further comprising, prior to the mounting:
 - forming the hole through the circuit board.
14. The method according to claim 6, wherein the air flow sensor and/or pressure sensor is a differential pressure sensor configured to measure a pressure difference between a pressure on the first side of the pressure sensor and a pressure on the second side of the pressure sensor.
15. The method according to claim 6, wherein the air flow sensor and/or pressure sensor is a micro electro-mechanical system (MEMS) sensor.
16. The method according to claim 6, the method further comprising:

providing a seal on an opposite side of the circuit board to the air flow sensor and/or pressure sensor to seal around the hole.

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